

So in a DISTINCT query output, each identical document appears only once. Note that all the SELECT-ed fields are taken into account. The query:

```
SELECT DISTINCT first_name, last_name FROM customers
```

returns -

Alice, Allen; Bob, Brown; Charlie, Clark; Alice, Brown; Bob, Clark; Charlie, Allen

So, first names and last names can repeat, but their pairs will not.

Technical restrictions to SELECT DISTINCT

When you have a query with DISTINCT and ORDER BY, you can only sort by a selected (visible) field:

```
SELECT DISTINCT first_name FROM customers ORDER BY first_name
```

- PASS

first_name which appears in ORDER BY must also appear in SELECT DISTINCT.

```
SELECT DISTINCT first_name FROM customers ORDER BY last_name
```

- FAILS

Compare the above to the regular:

```
SELECT first_name FROM customers ORDER BY last_name
```

- This works

The following might explain the behaviours of Distinct

1. SELECT DISTINCT returns only distinct values (different).
2. SELECT DISTINCT excludes from the reports of redundant information.
3. For aggregates: COUNT, AVG, MAX etc.
4. On single column DISTINCT works and not multiple columns.
5. Only with the SELECT statement is the DISTINCT clause used.
6. In the DISTINCT clause, you will be returning the unique values of the expression as you use just one expression.
7. The question can extract specific variations of the specified expressions if you bring more than one expression into the DISTINCT clause.
8. In MySQL, NULL values are not ignored by the DISTINCT clause.
9. There are also NULL as a different attribute if you use the DISTINCT clause in your SQL statement.

Another example for DISTINCT “With Single Expression” is finding the best example of the MySQL DISTINCT clause. We may use the MySQL DISTINCT clause to recover the duplication from a single field. Examples include:

```
SELECT DISTINCT state
```

```
FROM customers;
```